

Supporting Information

When High Accuracy Misleads in Literature-Derived Machine Learning for Deep Eutectic Solvent Recommendation

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Structure of this Supporting Information

This Supporting Information is organized into five sections:

1. Literature search and data extraction
2. Dataset construction and feature engineering
3. Modeling framework
4. Evaluation and robustness analyses
5. Reproducibility resources

Section S1 — Literature Search and Data Extraction

Text S1. Search Strategy and Study Selection

A structured literature search was conducted in Scopus to identify studies reporting the application of deep eutectic solvents (DESs) in pesticide extraction. The search covered publications available up to 30 December 2025 and was performed using the following query:

TITLE-ABS-KEY (("deep eutectic solvent*" OR "DES" OR "natural deep eutectic solvent*" OR "NADES") AND (pesticide* OR insecticide* OR herbicide* OR fungicide* OR agrochemical*) AND (extraction OR "sample preparation" OR microextraction))

The search yielded an initial set of 335 records. Document types were screened, and non-research documents were excluded, including review articles (n = 38), book chapters (n = 4), errata (n = 3), conference reviews (n = 2), and conference papers (n = 7). The remaining 281 research articles were subjected to eligibility screening.

Study selection was conducted in two stages. First, titles and abstracts were screened to identify potentially relevant studies. Second, full-text articles were evaluated for eligibility based on predefined inclusion and exclusion criteria.

Studies were included if:

- DES systems were used as extraction solvents in analytical workflows;
- extraction performance metrics (e.g., recovery) were reported at the experiment level;
- sufficient information was available to assign performance values to individual DES systems.

Studies were excluded if:

- DESs were used only as components of materials (e.g., sorbents or composites);
- extraction performance could not be attributed to individual DES systems;
- the study focused solely on physicochemical property prediction;
- experimental details were insufficient to reconstruct experiment-level records.
- Following full-text screening, 132 studies were retained for experiment-level data extraction.
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Text S2. AI-Assisted Experiment-Level Data Extraction and Validation

Extraction protocol

Experiment-level data were extracted using a structured prompt-based workflow implemented in NotebookLM, designed to capture all reported DES systems without aggregation or inference. Each record corresponded to a unique combination of DES composition (HBA, HBD, molar ratio), analyte identity, and reported performance metrics.

A strict zero-inference policy was applied: only explicitly reported values were recorded, and missing values were retained as “N/A”. Both optimal and suboptimal systems were included to preserve within-study comparability.

Manual validation and extraction audit

To assess extraction reliability, a random subset of 40 studies ($\approx 30\%$ of the dataset) was selected for manual validation against the original publications.

Validation was performed at the **field level**, focusing on the following variables:

- HBA identity
- HBD identity
- molar ratio
- analyte identity
- recovery (%)

A total of **1200 fields** were manually checked across the validated subset. Of these, **46 discrepancies** were identified prior to correction, corresponding to an extraction accuracy of **96.2% (field-level, pre-correction)**.

All identified discrepancies were subsequently corrected by direct comparison with the source publications.

Error characterization

Errors were not uniformly distributed across studies. Higher error rates were observed in:

- studies using non-standard or abbreviated component names for HBA/HBD,
- systems with fractional or non-integer molar ratios,
- multi-component DES systems (e.g., ternary mixtures).

These discrepancies primarily arose from challenges in parsing complex tabulated formats and inconsistent reporting conventions across studies.

Interpretation of validation results

The reported accuracy reflects **field-level agreement prior to manual correction** and should be interpreted as an estimate of extraction reliability under the applied protocol. After correction, all validated entries were aligned with the source data.

Given the heterogeneity of reporting formats and the uneven distribution of records across studies, this validation is best interpreted as an **audit of extraction consistency**, rather than a formal guarantee of error-free reconstruction. A quantitative summary of the field-level validation audit, including error counts and accuracy estimates, is provided in Table S1.

Table S1. Summary of extraction validation (field-level audit)

Metric	Value
Total studies included	132
Studies manually validated	40
Validation coverage	30%
Validation approach	Random sampling
Validation unit	Field-level audit
Fields evaluated	1200
Fields with discrepancies (pre-correction)	46
Extraction accuracy (pre-correction)	96.2%
Fields evaluated	HBA, HBD, molar ratio, analyte, recovery
Common error sources	Table parsing, name normalization, condition assignment
Higher-error cases	Non-standard component naming, fractional ratios, multi-component DES
Correction method	Manual verification against original publications
Post-correction status	All validated entries aligned with source data

Section S2 — Dataset and Feature Engineering

Text S3. Notebook LM Extraction Prompt

You are an elite, zero-hallucination AI Scientific Copilot specialized in Deep Eutectic Solvents (DES), Green Analytical Chemistry, and Experiment-Level Data Engineering.

Your mission is to extract a COMPLETE experiment-level dataset from DES-based pesticide/toxin extraction papers.

CRITICAL UPGRADE FROM v5.1:

You MUST extract ALL tested DES systems — not only the optimal one.

Do NOT collapse results to best-performing DES.

Extract sub-optimal, low-performing, and failed systems if reported.

I. NON-NEGOTIABLE EXTRACTION RULES

- 1) Extract ONLY explicitly reported experimental data.
- 2) NEVER infer missing values.
- 3) If a value is not explicitly stated → return "N/A".
- 4) If multiple DES systems are evaluated in tables or optimization sections:
 - Extract EACH row as a separate experiment entry.
 - Do NOT summarize.
 - Do NOT select only optimal conditions.
- 5) If a DES is reported as inferior or rejected, still extract it.
- 6) If recovery is reported for multiple analytes, extract each analyte as a separate entry.

II. PRIORITY TARGET

Focus ONLY on:

- DES used as extraction solvent

- Optimization tables comparing different DES systems
- Effect-of-DES-type studies
- Screening tables

If the paper is purely adsorption material (DES-modified composite) and does NOT report extraction recovery:

Return:

```
{
  "Article_Type": "Non-Extraction (Material Study)",
  "Reason": "No solvent extraction recovery data reported"
}
```

and STOP.

III. OUTPUT STRUCTURE (MANDATORY JSON FORMAT)

```
{
  "Bibliographic_Metadata": {
    "Article_Title": "",
    "Journal_Name": "",
    "DOI": "",
    "Publication_Year": "",
    "Extraction_Confidence_Score": 0-100
  },
  "Experiment_Level_Data": [
    {
      "DES_HBA_Name": "",
      "DES_HBD_Name": "",
      "Molar_Ratio": "",
      "Water_Content_Wt_Percent": float or "N/A",
      "Analyte_Name": "",
      "Analyte_Family": "",
      "Matrix_Category": "Water / Food / Soil / Biological / N/A",
      "Extraction_Technique": "",
      "Sample_pH": float or "N/A",
      "Extraction_Time_min": float or "N/A",
      "Temperature_C": float or "N/A",
      "Recovery_Percent": float or "N/A",
      "Enrichment_Factor": float or "N/A",
      "LOD_ug_per_L": float or "N/A",
      "Is_Optimal_System": true/false,
      "Was_Reported_As_Inferior": true/false,
      "Provenance": "Exact table/figure/section reference",
      "Confidence_Score": 0-100
    }
  ]
}
```

IV. SCREENING ENFORCEMENT LOGIC

For any table titled similar to:

- “Effect of DES type”
- “Optimization of DES”
- “Comparison of different DES”
- “Screening of DES systems”

You MUST:

- Extract ALL rows.

- Each DES system = one object inside Experiment_Level_Data.

- If recovery is lower than optimal → mark:

"Was_Reported_As_Inferior": true

- If text explicitly says "optimal DES" → mark:

"Is_Optimal_System": true

If not specified → set both to false.

V. IMPORTANT

DO NOT:

- Collapse multiple DES into one.

- Ignore low-recovery systems.

- Replace table values with text summary.

- Merge multiple analytes into one record.

If no screening table exists and only one DES tested:

Return single entry in Experiment_Level_Data.

VI. FINAL CHECK BEFORE OUTPUT

Before finalizing JSON:

Count how many DES systems were tested.

If more than one DES is mentioned in the paper and you output only one → YOU FAILED.

END OF PROTOCOL

Text S4. Interaction Feature Definitions

To represent relationships between DES systems and analytes, a set of engineered interaction features was constructed from interpretable molecular descriptors. These features were designed to capture both the internal composition of the DES and its physicochemical relationship to the analyte.

First, DES-level aggregate descriptors were computed from HBA and HBD component descriptors using a molar-ratio-weighted mean:

$$x_{\text{DES}}^{\text{wmean}} = \frac{n_{\text{HBA}} x_{\text{HBA}} + n_{\text{HBD}} x_{\text{HBD}}}{n_{\text{HBA}} + n_{\text{HBD}}}$$

where n_{HBA} and n_{HBD} are the parsed molar ratio coefficients of the hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD), respectively.

In addition, within-DES contrast descriptors were defined as absolute differences between HBA and HBD component values:

$$x_{\text{DES}}^{\text{absdiff}} = |x_{\text{HBA}} - x_{\text{HBD}}|$$

To characterize DES-analyte relationships, three generic transformations were applied for each descriptor family (molecular weight, logP, TPSA, hydrogen-bond acceptor count, hydrogen-bond donor count, rotatable bonds, ring count, and heavy atom count):

Absolute difference:

$$|x_{\text{analyte}} - x_{\text{DES}}^{\text{wmean}}|$$

Ratio:

$$\frac{x_{\text{analyte}}}{x_{\text{DES}}^{\text{wmean}}}$$

Product:

$$x_{\text{analyte}} \cdot x_{\text{DES}}^{\text{wmean}}$$

These transformations provide complementary views of compatibility. Absolute differences capture mismatch, ratios provide scale-sensitive comparisons, and products provide simple interaction-strength proxies.

Additional physically motivated proxy features were defined to approximate polarity and surface compatibility. A binary polarity-match feature was introduced as:

$$\mathbb{I}(|\log P_{\text{analyte}} - \log P_{\text{DES}}^{\text{wmean}}| \leq 1.5)$$

Similarly, a TPSA-match feature was defined as:

$$\mathbb{I}(|\text{TPSA}_{\text{analyte}} - \text{TPSA}_{\text{DES}}^{\text{wmean}}| \leq 20.0)$$

Hydrogen-bond complementarity was approximated using the following interaction terms:

$$\begin{aligned} &\text{HBA}_{\text{analyte}} \cdot \text{HBD}_{\text{DES}}^{\text{wmean}} \\ &\text{HBD}_{\text{analyte}} \cdot \text{HBA}_{\text{DES}}^{\text{wmean}} \end{aligned}$$

and their combined form:

$$(\text{HBA}_{\text{analyte}} \cdot \text{HBD}_{\text{DES}}^{\text{wmean}}) + (\text{HBD}_{\text{analyte}} \cdot \text{HBA}_{\text{DES}}^{\text{wmean}})$$

A hydrogen-bond balance mismatch feature was defined as:

$$|\text{HBA}_{\text{analyte}} - \text{HBD}_{\text{DES}}^{\text{wmean}}| + |\text{HBD}_{\text{analyte}} - \text{HBA}_{\text{DES}}^{\text{wmean}}|$$

Finally, molar-ratio-derived descriptors were included to preserve compositional information:

HBA fraction:

$$\frac{n_{\text{HBA}}}{n_{\text{HBA}} + n_{\text{HBD}}}$$

HBD fraction:

$$\frac{n_{\text{HBD}}}{n_{\text{HBA}} + n_{\text{HBD}}}$$

HBA-to-HBD ratio:

$$\frac{n_{\text{HBA}}}{n_{\text{HBD}}}$$

Together, these features provide a low-dimensional and interpretable representation of DES composition, internal contrast, and DES–analyte compatibility under leakage-clean conditions.

Table S2. Mathematical definitions of engineered features used to represent DES systems and DES–analyte interactions

Feature Type	Formula	Description
Weighted mean	$x_{DES}^{wmean} = \frac{n_{HBA}x_{HBA} + n_{HBD}x_{HBD}}{n_{HBA} + n_{HBD}}$	Composition-weighted descriptor reflecting DES composition, where n_{HBA} and n_{HBD} are the parsed molar ratio parts.
Absolute difference	$ x_1 - x_2 $	Magnitude of difference between two descriptor values
Ratio	$\frac{x_A}{x_B}$	Scale-sensitive proportional relationship between two descriptor values. Computed only when the denominator is non-zero.
Product	$x_A \times x_B$	Simple interaction-strength proxy based on simultaneous contribution of two descriptor values.
DES aggregate contrast	$ x_{HBA} - x_{HBD} $	Absolute difference between descriptor values of HBA and HBD components, representing internal compositional heterogeneity within the DES system.
DES–analyte absolute difference (Δ descriptor)	$\Delta x = x_{DES} - x_{analyte}$	Relative difference between DES and analyte properties
DES–analyte ratio	$\frac{x_{Analyte}}{x_{DES}^{wmean}}$	Relative proportional contrast between analyte and DES-level descriptor.
DES–analyte product	$x_{Analyte} \times x_{DES}^{wmean}$	Product-based interaction proxy between analyte and DES-level descriptor.
H-bond acceptor–donor product	$HBA_{Analyte} \times HBD_{DES}^{wmean}$	Proxy for complementarity between analyte hydrogen-bond acceptor capacity and DES donor capacity.
H-bond donor–acceptor product	$HBD_{Analyte} \times HBA_{DES}^{wmean}$	Proxy for complementarity between analyte hydrogen-bond donor capacity and DES acceptor capacity.
H-bond complement sum	$(HBA_{Analyte} \times HBD_{DES}^{wmean}) + (HBD_{Analyte} \times HBA_{DES}^{wmean})$	Aggregate proxy for total donor–acceptor complementarity.
H-bond balance	$HBD_{DES} - HBA_{analyte}$	Hydrogen-bond complementarity proxy
Polarity mismatch (binary)	$ \log P_{DES} - \log P_{analyte} $	Absolute difference in polarity (like-dissolves-like principle)
TPSA match (binary)	$\mathbb{1}(TPSA_{Analyte} - TPSA_{DES}^{wmean} \leq 20.0)$	Binary indicator of approximate polar surface compatibility between analyte and DES, defined using a fixed threshold on the difference in topological polar surface area (TPSA).
Ratio fraction (HBA)	$\frac{n_{HBA}}{n_{HBA} + n_{HBD}}$	Fractional contribution of HBA in the parsed binary DES molar ratio.
Ratio fraction (HBD)	$\frac{n_{HBD}}{n_{HBA} + n_{HBD}}$	Fractional contribution of HBD in the parsed binary DES molar ratio.
HBA-to-HBD ratio	$\frac{n_{HBA}}{n_{HBD}}$	Direct molar ratio between HBA and HBD parts.

Table S3. Molecular descriptors and derived features used for model development. Descriptors were computed from canonical SMILES representations using RDKit and include component-level descriptors, DES-level aggregates, and DES-analyte interaction features.

Feature	Description
analyte_hba	Hydrogen bond acceptor count of the analyte
analyte_hbd	Hydrogen bond donor count of the analyte
analyte_heavyatoms	Number of heavy atoms of the analyte
analyte_logp	Octanol–water partition coefficient (logP) of the analyte
analyte_molwt	Molecular weight of the analyte
analyte_ringcount	Number of rings of the analyte
analyte_rotbonds	Number of rotatable bonds of the analyte
analyte_tpsa	Topological polar surface area (TPSA) of the analyte
des_hba_absdiff	Absolute difference in hydrogen bond acceptor count between HBA and HBD components (DES internal contrast)
des_hba_wmean	Molar-ratio-weighted mean of hydrogen bond acceptor count for the DES
des_hbd_absdiff	Absolute difference in hydrogen bond donor count between HBA and HBD components
des_hbd_wmean	Molar-ratio-weighted mean of hydrogen bond donor count for the DES
des_logp_absdiff	Absolute difference in logP between HBA and HBD components
des_logp_wmean	Molar-ratio-weighted mean of logP for the DES
des_molwt_absdiff	Absolute difference in molecular weight between HBA and HBD components
des_molwt_wmean	Molar-ratio-weighted mean of molecular weight for the DES
des_ringcount_absdiff	Absolute difference in ring count between HBA and HBD components
des_ringcount_wmean	Molar-ratio-weighted mean of ring count for the DES
des_rotbonds_absdiff	Absolute difference in number of rotatable bonds between HBA and HBD components
des_rotbonds_wmean	Molar-ratio-weighted mean of number of rotatable bonds for the DES
des_tpsa_absdiff	Absolute difference in TPSA between HBA and HBD components
des_tpsa_wmean	Molar-ratio-weighted mean of TPSA for the DES
hba_hba	Hydrogen bond acceptor count of the HBA component
hba_hbd	Hydrogen bond donor count of the HBA component
hba_logp	LogP of the HBA component
hba_molwt	Molecular weight of the HBA component
hba_ringcount	Number of rings of the HBA component
hba_rotbonds	Number of rotatable bonds of the HBA component
hba_tpsa	TPSA of the HBA component
hbd_hba	Hydrogen bond acceptor count of the HBD component
hbd_hbd	Hydrogen bond donor count of the HBD component
hbd_logp	LogP of the HBD component
hbd_molwt	Molecular weight of the HBD component

hbd_ringcount	Number of rings of the HBD component
hbd_rotbonds	Number of rotatable bonds of the HBD component
hbd_tpsa	TPSA of the HBD component
pair_hba_absdiff	Absolute difference between analyte and DES-level hydrogen bond acceptor count
pair_hba_product	Product of analyte and DES-level hydrogen bond acceptor count
pair_hba_ratio	Ratio between analyte and DES-level hydrogen bond acceptor count
pair_hbd_absdiff	Absolute difference between analyte and DES-level hydrogen bond donor count
pair_hbd_product	Product of analyte and DES-level hydrogen bond donor count
pair_hbd_ratio	Ratio between analyte and DES-level hydrogen bond donor count
pair_hbond_acceptor_donor_product	Product of analyte HBA and DES donor capacity (HBD)
pair_hbond_donor_acceptor_product	Product of analyte HBD and DES acceptor capacity (HBA)
pair_hbond_complement_sum	Combined hydrogen-bond complementarity between analyte and DES
pair_hbond_balance_absdiff	Hydrogen-bond balance mismatch between analyte and DES
pair_heavyatoms_absdiff	Absolute difference between analyte and DES-level heavy atom count
pair_heavyatoms_product	Product of analyte and DES-level heavy atom count
pair_heavyatoms_ratio	Ratio between analyte and DES-level heavy atom count
pair_logp_absdiff	Absolute difference between analyte and DES-level logP
pair_logp_match_binary	Binary indicator of approximate polarity match between analyte and DES
pair_logp_product	Product of analyte and DES-level logP
pair_logp_ratio	Ratio between analyte and DES-level logP
pair_molwt_absdiff	Absolute difference between analyte and DES-level molecular weight
pair_molwt_product	Product of analyte and DES-level molecular weight
pair_molwt_ratio	Ratio between analyte and DES-level molecular weight
pair_ringcount_absdiff	Absolute difference between analyte and DES-level ring count
pair_ringcount_product	Product of analyte and DES-level ring count
pair_rotbonds_absdiff	Absolute difference between analyte and DES-level rotatable bonds
pair_rotbonds_product	Product of analyte and DES-level rotatable bonds
pair_rotbonds_ratio	Ratio between analyte and DES-level rotatable bonds
pair_tpsa_absdiff	Absolute difference between analyte and DES-level TPSA
pair_tpsa_match_binary	Binary indicator of approximate TPSA match between analyte and DES
pair_tpsa_product	Product of analyte and DES-level TPSA
pair_tpsa_ratio	Ratio between analyte and DES-level TPSA
ratio_hba_fraction	Fractional contribution of HBA in the DES molar ratio

ratio_hba_part	Parsed HBA coefficient in the DES molar ratio
ratio_hba_to_hbd	HBA-to-HBD molar ratio
ratio_hbd_part	Parsed HBD coefficient in the DES molar ratio

Table S4. Features excluded during preprocessing and the corresponding reasons for removal, including leakage risk, low variance, or missing values.

Feature name	Missing fraction train	Drop reason stage	Selected for stage5
ratio_hba_part	0	keep_after_missingness	TRUE
ratio_hbd_part	0	keep_after_missingness	TRUE
ratio_hba_fraction	0	keep_after_missingness	TRUE
ratio_hbd_fraction	0	drop_high_correlation	FALSE
ratio_hba_to_hbd	0	keep_after_missingness	TRUE
analyte_hba	0	keep_after_missingness	TRUE
analyte_hbd	0	keep_after_missingness	TRUE
analyte_heavyatoms	0	keep_after_missingness	TRUE
analyte_logp	0	keep_after_missingness	TRUE
analyte_molwt	0	keep_after_missingness	TRUE
analyte_ringcount	0	keep_after_missingness	TRUE
analyte_rotbonds	0	keep_after_missingness	TRUE
analyte_tpsa	0	keep_after_missingness	TRUE
des_hba_absdiff	0	keep_after_missingness	TRUE
des_hba_wmean	0	keep_after_missingness	TRUE
des_hbd_absdiff	0	keep_after_missingness	TRUE
des_hbd_wmean	0	keep_after_missingness	TRUE
des_heavyatoms_absdiff	0	drop_high_correlation	FALSE
des_heavyatoms_wmean	0	drop_high_correlation	FALSE
des_logp_absdiff	0	keep_after_missingness	TRUE
des_logp_wmean	0	keep_after_missingness	TRUE
des_molwt_absdiff	0	keep_after_missingness	TRUE
des_molwt_wmean	0	keep_after_missingness	TRUE
des_ringcount_absdiff	0	keep_after_missingness	TRUE
des_ringcount_wmean	0	keep_after_missingness	TRUE
des_rotbonds_absdiff	0	keep_after_missingness	TRUE
des_rotbonds_wmean	0	keep_after_missingness	TRUE
des_tpsa_absdiff	0	keep_after_missingness	TRUE
des_tpsa_wmean	0	keep_after_missingness	TRUE
hba_descriptor_eligible_phase1	0	drop_constant	FALSE
hba_hba	0	keep_after_missingness	TRUE
hba_hbd	0	keep_after_missingness	TRUE
hba_heavyatoms	0	drop_high_correlation	FALSE
hba_logp	0	keep_after_missingness	TRUE
hba_molwt	0	keep_after_missingness	TRUE
hba_ringcount	0	keep_after_missingness	TRUE

hba_rotbonds	0	keep_after_missingness	TRUE
hba_tpsa	0	keep_after_missingness	TRUE
hbd_descriptor_eligible_phase1	0	drop_constant	FALSE
hbd_hba	0	keep_after_missingness	TRUE
hbd_hbd	0	keep_after_missingness	TRUE
hbd_heavyatoms	0	drop_high_correlation	FALSE
hbd_logp	0	keep_after_missingness	TRUE
hbd_molwt	0	keep_after_missingness	TRUE
hbd_ringcount	0	keep_after_missingness	TRUE
hbd_rotbonds	0	keep_after_missingness	TRUE
hbd_tpsa	0	keep_after_missingness	TRUE
pair_hba_absdiff	0	keep_after_missingness	TRUE
pair_hba_product	0	keep_after_missingness	TRUE
pair_hba_ratio	0	keep_after_missingness	TRUE
pair_hbd_absdiff	0	keep_after_missingness	TRUE
pair_hbd_product	0	keep_after_missingness	TRUE
pair_hbd_ratio	0	keep_after_missingness	TRUE
pair_hbond_acceptor_donor_product	0	keep_after_missingness	TRUE
pair_hbond_balance_absdiff	0	keep_after_missingness	TRUE
pair_hbond_complement_sum	0	keep_after_missingness	TRUE
pair_hbond_donor_acceptor_product	0	keep_after_missingness	TRUE
pair_heavyatoms_absdiff	0	keep_after_missingness	TRUE
pair_heavyatoms_product	0	keep_after_missingness	TRUE
pair_heavyatoms_ratio	0	keep_after_missingness	TRUE
pair_logp_absdiff	0	keep_after_missingness	TRUE
pair_logp_match_binary	0	keep_after_missingness	TRUE
pair_logp_product	0	keep_after_missingness	TRUE
pair_logp_ratio	0	keep_after_missingness	TRUE
pair_molwt_absdiff	0	keep_after_missingness	TRUE
pair_molwt_product	0	keep_after_missingness	TRUE
pair_molwt_ratio	0	keep_after_missingness	TRUE
pair_ringcount_absdiff	0	keep_after_missingness	TRUE
pair_ringcount_product	0	keep_after_missingness	TRUE
pair_ringcount_ratio	0.326014	drop_high_missingness	FALSE
pair_rotbonds_absdiff	0	keep_after_missingness	TRUE
pair_rotbonds_product	0	keep_after_missingness	TRUE
pair_rotbonds_ratio	0	keep_after_missingness	TRUE
pair_tpsa_absdiff	0	keep_after_missingness	TRUE
pair_tpsa_match_binary	0	keep_after_missingness	TRUE
pair_tpsa_product	0	keep_after_missingness	TRUE
pair_tpsa_ratio	0	keep_after_missingness	TRUE
ratio_parse_ok	0	drop_constant	FALSE

Section S3 — Modeling Framework

Text S5. Mathematical Formulation of the Hybrid Model

The hybrid model combines a global classification component with a pairwise preference component to produce a ranking of candidate DES systems within each (DOI, analyte) group. Let x_i denote the feature vector corresponding to DES–analyte record i . The global classification model estimates the probability that a given DES system is optimal within its competitive group:

$$P(y_i = 1 \mid x_i)$$

where $y_i = 1$ indicates that the DES system is labeled as optimal within its (DOI, analyte) group. To model relative preferences between candidate systems, a pairwise formulation is introduced. For a pair of DES systems (i, j) within the same group, the pairwise feature vector is defined as:

$$\Delta x_{ij} = x_i - x_j$$

The pairwise model produces a directional preference score:

$$f(\Delta x_{ij})$$

which is converted to a probability using a sigmoid transformation:

$$p_{ij} = \frac{1}{1 + \exp(-f(\Delta x_{ij}))}$$

where p_{ij} represents the probability that system i is preferred over system j .

For each (DOI, analyte) group containing n candidate DES systems, pairwise comparisons are performed between all pairs, and the pairwise scores are aggregated to obtain a group-level preference score:

$$S_i^{pair} = \frac{1}{n-1} \sum_{j \neq i} p_{ij}$$

The final hybrid score is defined as a weighted combination of the global classification probability and the aggregated pairwise score:

$$S_i^{final} = \alpha_{full} \cdot P(y_i = 1 \mid x_i) + \alpha_{pair} \cdot S_i^{pair}$$

subject to:

$$\alpha_{full} + \alpha_{pair} = 1$$

In this study, fixed weights of $\alpha_{full} = 0.7$ and $\alpha_{pair} = 0.3$ were used to prioritize the stability of the global classification component while incorporating complementary comparative information. Importantly, all components of the hybrid model are constructed using leakage-clean features, and pairwise inference is performed exclusively within each evaluation group. No outcome-derived variables are used in either component.

Given the limited number of informative (DOI, analyte) groups and the high redundancy of pairwise comparisons (Table S6), the pairwise component should be interpreted as providing complementary but structurally constrained information. Under these conditions, the hybrid formulation is not expected to fully resolve ranking limitations arising from the underlying dataset structure.

Text S6. Pairwise Dataset Structure and Statistics

The leakage-clean pairwise dataset used in this study was constructed exclusively from informative (DOI, analyte) groups identified within the training split of the final rebuilt pipeline, rather than from the full dataset.

In the final leakage-clean training set, 74 directional pairwise samples were generated from 11 informative groups, corresponding to 37 effective unordered comparisons under bidirectional augmentation. The number of directionals pairwise samples per group ranged from 4 to 24, with a median of 4.

Although the nominal number of pairwise samples is non-zero, the effective diversity of the dataset remains limited. All pairwise comparisons are generated within a small number of groups, leading to repeated reuse of the same candidate systems within groups and substantial within-group redundancy.

Importantly, these training-only pairwise statistics should not be conflated with the full-dataset informative-group count reported in Section 3.1. The latter describes the structural sparsity of informative competition across the complete reconstructed dataset, whereas the pairwise dataset reported here reflects only the subset of informative groups available after split-first reconstruction and training-only pairwise construction.

Taken together, these characteristics indicate that the leakage-clean pairwise training dataset is small, highly structured, and underpowered for learning robust transferable preference relationships across studies.

Table S5. Hyperparameter settings used for model training. Conservative configurations were selected to reduce overfitting given the limited size and structure of the dataset.

Model	Parameter	Value
Logistic Regression (l2)	penalty	l2
Logistic Regression (l2)	C	1
Logistic Regression (l2)	class_weight	balanced
Logistic Regression (l2)	solver	lbfgs
Logistic Regression (l2)	max_iter	5000

Logistic Regression (elasticnet)	penalty	elasticnet
Logistic Regression (elasticnet)	l1_ratio	0.5
Logistic Regression (elasticnet)	C	0.5
Logistic Regression (elasticnet)	class_weight	balanced
Logistic Regression (elasticnet)	solver	saga
Logistic Regression (elasticnet)	max_iter	8000
Random Forest	n_estimators	500
Random Forest	class_weight	balanced_subsample
Random Forest	random_state	42
XGBoost	n_estimators	400
XGBoost	max_depth	4
XGBoost	learning_rate	0.05
XGBoost	subsample	0.9
XGBoost	colsample_bytree	0.9
XGBoost	reg_lambda	1
XGBoost	objective	binary:logistic
XGBoost	eval_metric	logloss
Hybrid model	alpha_full	0.7
Hybrid model	alpha_pair	0.3

Table S6. Quantitative summary of the leakage-clean pairwise dataset structure

Metric	Value
Total pairwise samples	74
Unique informative groups (DOI, analyte)	11
Pair samples per group (median)	4
Pair samples per group (range)	4–24
Label distribution (preferred vs non-preferred)	balanced (by construction)
Pair construction method	bidirectional ($i \rightarrow j$ and $j \rightarrow i$)
Effective independent comparisons	37 (unique unordered pairs)
Redundancy structure	high (within-group reuse of candidates)
Informative vs total groups	11 training-informative groups used for pairwise construction; distinct from the 21 full-dataset informative groups reported in Section 3.1

Section S4 — Evaluation and Robustness Analyses

Text S7. Additional robustness analyses (Ablation and sensitivity)

- **Ablation study of the hybrid formulation**

Ablation analysis was performed to isolate the contributions of the full-model and pairwise components to ranking performance. Results show that the pairwise-only formulation performs substantially worse than the full-only model, and that the hybrid model does not improve over the full component alone. These analyses were conducted on the comparable-group subset under leakage-clean conditions and are consistent with, but not identical to, the final holdout evaluation reported in Tables S7–S9. The results therefore support the conclusion that the pairwise signal does not contribute stable additional ranking value under the present dataset structure (Table S12).

- **Sensitivity of ranking performance to the hybrid weighting parameter**

Sensitivity analysis was performed by varying the hybrid weighting parameter α_{full} while evaluating ranking performance on comparable groups with at least two candidates. These analyses were conducted on the same restricted evaluation subset and should be interpreted as exploratory diagnostics rather than direct estimates of final holdout performance. The results indicate that the lack of hybrid improvement is not attributable to a specific weight choice, but reflects a structural limitation of the pairwise signal (Table S13).

Text S8. Split-wise preprocessing, pairwise construction, and inference (pseudocode)

Below is a step-wise description of the leakage-clean pipeline applied in this study:

Step 1 — Grouped data splitting

- Partition dataset by DOI
- Assign all records from each DOI to either training or test set

Step 2 — Training-only preprocessing

- Fit preprocessing steps (missing-value filtering, variance filtering, collinearity filtering) using training data only
- Record selected feature set

Step 3 — Transformation of test data

- Apply the preprocessing transformations derived from training data to the test set
- No re-fitting is performed on test data

Step 4 — Pairwise dataset construction (training only)

- Identify informative (DOI, analyte) groups within the training set
- Construct pairwise samples only within these training groups

- Apply bidirectional construction ($i \rightarrow j$ and $j \rightarrow i$)

Step 5 — Model training

- Train full (global classification) model on training records
- Train pairwise model on training pairwise dataset

Step 6 — Inference on held-out groups

- For each test (DOI, analyte) group:
 - Compute full-model probabilities for all candidates
 - Perform pairwise comparisons within the same test group
 - Aggregate pairwise scores at group level

Step 7 — Ranking and evaluation

- Rank candidates within each test group using model scores
- Compute group-level ranking metrics
- Perform restricted evaluation on comparable multi-candidate groups

This procedure ensures strict separation between training and test data and prevents leakage from outcome-derived or test-set information into model construction.

Table S7. Ranking performance on all groups and comparable multi-candidate groups under leakage-clean DOI-held-out evaluation.

Scope	N groups	Baseline Hit@1	Hybrid Hit@1	Baseline Hit@3	Hybrid Hit@3	Baseline MRR	Hybrid MRR	Hybrid better	Hybrid worse	Equal
All groups in predictions	119	0.983	0.975	0.991	0.991	0.987	0.980	0	3	116
Comparable groups ≥ 2	8	0.75	0.625	0.875	0.875	0.802	0.702	0	3	5
Comparable groups ≥ 3	6	0.667	0.50	0.833	0.833	0.736	0.603	0	3	3

Table S8. Ranking performance with uncertainty under leakage-clean DOI-held-out evaluation

Scope	N groups	Model	Hit@1	95% CI (Wilson)	MRR	95% CI (Bootstrap)
All groups	119	Baseline	0.983	0.9580–1.0000	0.987	0.9657–1.0000
All groups	119	Hybrid	0.958	0.9412–1.0000	0.980	0.9544–1.0000
Comparable groups ≥ 2	8	Baseline	0.750	0.3750–1.0000	0.802	0.5208–1.0000
Comparable groups ≥ 2	8	Hybrid	0.625	0.2500–1.0000	0.702	0.4280–1.0000
Comparable groups ≥ 3	6	Baseline	0.667	0.3292–1.0000	0.736	0.4224–1.0000
Comparable groups ≥ 3	6	Hybrid	0.500	0.1667–0.8333	0.603	0.2857–0.8889

Table S9. Group-wise statistical comparison.

Scope	n	Hybrid better	Hybrid worse	p-value
≥ 2 candidates	3	0	3	0.25
≥ 3 candidates	3	0	3	0.25

Text S9. Group-wise statistical comparison

An exact sign (binomial) test was applied to group-wise comparison outcomes derived from Table S9, with tied cases excluded.

For comparable groups with ≥ 2 candidates ($n = 3$), the hybrid model did not outperform the baseline in any group, and was worse in 3 groups ($p = 0.25$). The same result was observed for groups with ≥ 3 candidates ($n = 3$), where the hybrid model again showed 0 wins and 3 losses relative to the baseline ($p = 0.25$).

These results confirm that no statistically robust evidence supports hybrid superiority under the final leakage-clean evaluation. Instead, the observed pattern is consistent with the broader conclusion of this study: under sparse comparative supervision and limited comparable-group sample size, pairwise augmentation does not provide stable ranking benefit beyond the baseline model.

Table S10. Detailed cross-validation results across folds for baseline models. Metrics are reported for each fold to illustrate variability and robustness of model performance.

Model	No. of features	ROC-AUC (mean)	ROC-AUC (SD)	PR-AUC (mean)	PR-AUC (SD)	MCC (mean)	MCC (SD)	Balanced accuracy (mean)	Balanced accuracy (SD)
logreg	75	0.568881	0.168497	0.926342	0.05797	0.0238	0.109799	0.496437	0.097752
xgb	75	0.565677	0.100964	0.926287	0.047015	-0.00872	0.064539	0.484506	0.039424
rf	75	0.5845	0.113426	0.934727	0.044276	-0.04043	0.023996	0.491309	0.005982

Table S11. Impact of validation strategy on baseline model performance

Validation Strategy	ROC-AUC	PR-AUC	MCC	Balanced Accuracy
Random row-wise split	0.983	0.997	0.794	0.889
DOI-grouped split	0.802	0.868	0.629	0.779

Table S12. Ablation analysis of the hybrid model under leakage-clean DOI-held-out evaluation.

Scope	Model	N groups	Hit@1	MRR
All groups	Full-only	119	0.9748	0.9807
All groups	Pairwise-only	119	0.9412	0.9614
All groups	Hybrid	119	0.9580	0.9702
Comparable ≥ 2	Full-only	8	0.6250	0.7125
Comparable ≥ 2	Pairwise-only	8	0.1250	0.4262
Comparable ≥ 2	Hybrid	0.3750	0.5565	
Comparable ≥ 3	Full-only	6	0.5000	0.6167
Comparable ≥ 3	Pairwise-only	6	0.1667	0.4016
Comparable ≥ 3	Hybrid	6	0.3333	0.4921

Table S13. Sensitivity of ranking performance to the hybrid weighting parameter (α) under comparable-group evaluation

α _full	α _pair	N groups	Hit@1	MRR
0.0	1.0	8	0.125	0.4262
0.3	0.7	8	0.125	0.4251
0.5	0.5	8	0.375	0.5543
0.7	0.3	8	0.375	0.5565
1.0	0.0	8	0.625	0.7125

Section S5 — Reproducibility Resources

Table S14. Software environment and libraries used for data processing and modeling.

Component	Version
Python	3.12.13
pandas	2.2.2
numpy	2.0.2
scikit-learn	1.6.1
matplotlib	3.10.0
RDKit	Not available in runtime environment (descriptors precomputed)
XGBoost	3.2.0

Table S15. Summary of the leakage-clean feature space used for modeling, grouped by descriptor family and functional role.

Feature family	N features	Representative examples	Functional role
Analyte descriptors	8	analyte_hba, analyte_hbd, analyte_heavyatoms	Physicochemical properties of the analyte
DES-level descriptors	16	des_hba_absdiff, des_hba_wmean, des_hbd_absdiff	Composition-weighted and contrast descriptors of the DES system
HBA descriptors	8	hba_hba, hba_hbd, hba_heavyatoms	Physicochemical properties of the hydrogen-bond acceptor
HBD descriptors	8	hbd_hba, hbd_hbd, hbd_heavyatoms	Physicochemical properties of the hydrogen-bond donor
DES-analyte interaction features	30	pair_hba_absdiff, pair_hba_product, pair_hba_ratio	Relative compatibility and interaction proxies between DES and analyte
Composition / ratio-derived features	5	ratio_hba_fraction, ratio_hba_part, ratio_hba_to_hbd	Descriptors derived from HBA/HBD composition and molar ratio

Table S16. Leakage-clean DOI-held-out classification performance of baseline and hybrid models.

Model	ROC-AUC	PR-AUC	MCC	Balanced Accuracy	Precision	Recall	F1-score
Baseline (LogReg) full clean	0.640648	0.749764	0.21471	0.604167	0.782258	0.808333	0.795082
Hybrid true pairwise clean	0.694444	0.787576	0.327805	0.675	0.833333	0.75	0.789474

Text S10. Sanity Baselines for Group-Level Ranking

Two sanity baselines were evaluated to contextualize group-level ranking performance under the leakage-clean DOI-held-out setting.

The first baseline consists of repeated random ranking within each (DOI, analyte) group, averaged over multiple independent permutations to estimate expected performance in the absence of learned structure. This baseline provides a lower-bound reference for evaluating whether model performance exceeds trivial ranking behavior.

The second baseline is a deterministic heuristic based on analyte–DES polarity mismatch, operationalized as the absolute difference between analyte logP and the DES-level weighted-mean logP. Tie-breaking was performed deterministically using TPSA mismatch, hydrogen-bond balance mismatch, and DES system identifier to ensure reproducibility.

Both baselines were evaluated using the same group structure, ranking metrics (Hit@1, Hit@3, MRR), and restricted comparable-group subsets as used for the primary model comparison.

The results show that while machine learning models outperform random ranking, their advantage over simple heuristic baselines is limited in comparable-group evaluation. This finding supports the interpretation that a substantial fraction of the observed ranking performance can be explained by coarse compatibility patterns already captured in the descriptor space, rather than by robust learning of transferable ranking relationships.

Table S17. Sanity baseline comparison under leakage-clean DOI-held-out evaluation.

Scope	n-groups	Random Hit@1 (mean $\pm$ SD)	Random Hit@3 (mean $\pm$ SD)	Random MRR (mean $\pm$ SD)	Heuristic Hit@1	Heuristic Hit@3	Heuristic MRR	Baseline Hit@1	Baseline Hit@3	Baseline MRR	Hybrid Hit@1	Hybrid Hit@3	Hybrid MRR
All groups	119	0.951 $\pm$ 0.010	0.978 $\pm$ 0.008	0.967 $\pm$ 0.006	0.950	0.983	0.966	0.983	0.983	0.987	0.975	0.983	0.980
≥ 2 candidates	8	0.273 $\pm$ 0.147	0.674 $\pm$ 0.125	0.507 $\pm$ 0.095	0.250	0.750	0.489	0.750	0.750	0.802	0.625	0.750	0.702
≥ 3 candidates	6	0.193 $\pm$ 0.156	0.566 $\pm$ 0.167	0.423 $\pm$ 0.113	0.167	0.667	0.402	0.667	0.667	0.736	0.500	0.667	0.603

Detailed sanity baseline comparison under leakage-clean DOI-held-out evaluation. Random baseline metrics are reported as mean $\pm$ standard deviation across repeated permutations. The heuristic baseline is deterministic and based on analyte–DES polarity mismatch with fixed tie-breaking rules. Results confirm that while machine learning models outperform random ranking, their advantage over simple heuristic rules is limited in comparable-group evaluation, supporting the interpretation that ranking performance is strongly influenced by coarse descriptor-level compatibility patterns.

Dataset S1

Code S1. Reproducible Leakage-Clean Modeling Pipeline

Code S1 provides the complete computational workflow for leakage-clean data processing, model training, and evaluation under the split-first, train-only paradigm used in this study. The pipeline includes DOI-grouped data splitting, training-only preprocessing, construction of the pairwise dataset exclusively from training groups, and test-only inference with within-group ranking evaluation.

The code reproduces all results reported in the manuscript and Supporting Information, including classification metrics, group-level ranking performance, uncertainty estimation, and statistical comparison. It ensures consistency of the implemented workflow with the leakage-control and evaluation principles described in Sections 2.4–2.7 and Text S8.

While the code guarantees reproducibility of the reported results, it does not constitute a formal proof of the absence of all possible indirect dependencies beyond the explicitly controlled pipeline steps.

The workflow additionally includes sanity baseline evaluation using repeated random ranking and a deterministic physicochemical heuristic, enabling direct comparison of learned models with trivial and rule-based alternatives.

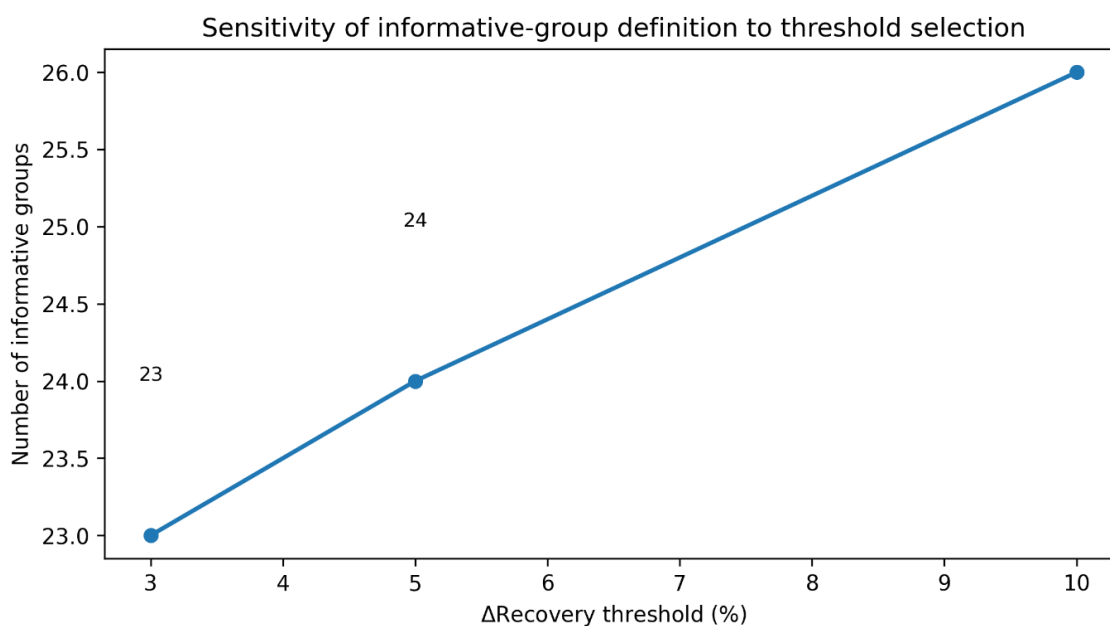


Figure S1. Sensitivity of the number of informative competitive groups to the Δ Recovery threshold used for group definition. The overall conclusion remains stable across reasonable threshold choices, indicating that the scarcity of informative groups is not an artifact of the 5% threshold.